

OUTPUT

```
Python 3.14.2 (tags/v3.14.2:dff79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/mithr/OneDrive/Desktop/PYTHON_CODE.py =====

===== ARTIFICIAL INTELLIGENCE =====
1. Water Jug Problem
2. Mars Rover Agent
3. 8 Queens Problem
4. OLA Cab Booking Agent
5. Uniform Cost Search (UCS)
6. Exit

Enter your Choice: 1

Water Jug Problem Solution
Step 1: (0,0) -> (0,3)
Step 2: (0,3) -> (3,0)
Step 3: (3,0) -> (3,3)
Step 4: (3,3) -> (4,2)
Step 5: (4,2) -> (0,2)
Step 6: (0,2) -> (2,0)

Goal Achieved!
4-Gallon Jug contains exactly 2 Gallons.

===== ARTIFICIAL INTELLIGENCE =====
1. Water Jug Problem
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6. Exit

Enter your Choice: 2

Mars Rover Agent

Percepts
- Camera Images
- Rock Samples
- Soil Samples
- Temperature
- Obstacles
- Battery Level

Actions

Ln: 126 Col: 19

===== ARTIFICIAL INTELLIGENCE =====
1. Water Jug Problem
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6. Exit

Enter your Choice: 3

8 Queens Solution

[1, 0, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 1, 0]
[0, 0, 0, 0, 1, 0, 0, 0]
[0, 0, 0, 0, 0, 0, 0, 1]
[0, 1, 0, 0, 0, 0, 0, 0]
[0, 0, 0, 1, 0, 0, 0, 0]
[0, 0, 0, 0, 0, 1, 0, 0]
[0, 0, 1, 0, 0, 0, 0, 0]

===== ARTIFICIAL INTELLIGENCE =====
1. Water Jug Problem
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6. Exit

Enter your Choice: 4
Enter Pickup Location : vellore
Enter Destination : ponnammalle

Available Cab Types
1. Mini

Ln: 126 Col: 19
```

```
*IDLE Shell 3.14.2*
File Edit Shell Debug Options Window Help

1. Water Jug Problem
2. Mars Rover Agent
3. 8 Queens Problem
4. OLA Cab Booking Agent
5. Uniform Cost Search (UCS)
6. Exit

Enter your Choice: 4
Enter Pickup Location : vellore
Enter Destination : ponnamalle

Available Cab Types
1. Mini
2. Micro
3. Sedan
4. Prime
5. Shared
Select Cab Type : prime

Cab Booked Successfully
Pickup : vellore
Destination : ponnamalle
Driver Assigned
Happy Journey!

===== ARTIFICIAL INTELLIGENCE =====
1. Water Jug Problem
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Enter your Choice: 5

Least Cost Path : S -> B -> C -> G
Total Cost : 8

===== ARTIFICIAL INTELLIGENCE =====
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5. Uniform Cost Search (UCS)
6. Exit

Enter your Choice: |
```